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Guihua Wang

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The Effect of Medicaid Expansion on Wait Time in the Emergency Department

Guihua Wang[a]

[a] Jindal School of Management, University of Texas at Dallas, Richardson, Texas 75080

Contact: guihua.wang@utdallas.edu, https://orcid.org/0000-0001-8324-7073 (GW)

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Abstract. We study the effect of Medicaid expansion on wait time in the emergency department (ED), using a difference-in-differences approach, where the treatment group includes the states that expanded Medicaid at the beginning of 2014 and the control group includes the states that did not expand Medicaid before 2016. We first focus on the average treatment effect and find that Medicaid expansion increases ED wait time by 10.4% (or 3.5 minutes). We then develop a first-difference causal forest (FDCF) approach for heterogeneous treatment-effect analysis by incorporating the first-difference approach into the causal forest approach. Our comprehensive simulation studies show that the FDCF approach has smaller mean-squared errors than direct applications of the causal forest approach. Finally, we apply the FDCF approach to our empirical example and find that the effect of Medicaid expansion varies widely across different hospitals. Our results are useful to policymakers deciding whether to expand Medicaid and to hospital managers seeking to understand the effect of Medicaid expansion on their hospitals. Our results are also useful to researchers applying causal machine learning methods for heterogeneous treatment-effect analysis.

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1. Introduction

Medicaid is a government insurance program that helps patients with limited income and resources with their medical costs. This program is jointly funded by state and federal governments and managed by states. Coverage of Medicaid varies by state, but is generally limited to low-income children, women, older adults, and people with disabilities who receive Supplemental Security Income. No significant changes had been made to the Medicaid program over the past few decades until the passage of the Affordable Care Act (sometimes known as the ACA, PPACA, or "Obamacare") in 2010. The ACA provided individual states an option to expand Medicaid coverage for uninsured adults and children with incomes below 138% of the federal poverty level (Kaiser Family Foundation 2013a).

The pros and cons of the Medicaid expansion program have been the subject of ongoing debate. Whereas proponents believe the expansion increases access to care for poor patients and reduces the financial burden on hospitals, opponents argue that the expansion puts extra financial burdens on other taxpayers, strains existing healthcare systems, and increases crowding of hospitals (Ku et al. 2011). Because the overall benefit of Medicaid expansion is unclear, states have made different decisions regarding the expansion. In 2014, a total of 26 states chose to expand Medicaid. Since 2014, an additional 10 states have implemented expansions, although 14 states still have not implemented expansions, as of 2021 (Kaiser Family Foundation 2021a).

The effect of Medicaid expansion on emergency department (ED) crowding has been of interest to policymakers and hospital managers for two reasons. First, EDs are more expensive than many other departments in a hospital. For example, treating a nonurgent condition in an ED costs two to five times more than treating the same condition in a primary care setting (New England Healthcare Institute 2010). Second, EDs are the most challenging department with respect to patient delays. ED crowding increases seven-day mortality by an estimated 30% (Sprivulis et al. 2006). Findings on the effect of Medicaid expansion on ED crowding are mixed. Whereas some studies find that Medicaid expansion increases ED crowding (see, e.g., Dresden et al. 2017 and Nikpay et al. 2017), others find that Medicaid expansion reduces or does not affect ED crowding (see, e.g., Pines et al. 2016, Klein et al. 2017, Sabik et al. 2017).

The discrepancies between existing studies suggest that one needs to look at not only the average effect, but also the heterogeneous effects of Medicaid expansion, because the effect may be positive for some hospitals and negative or insignificant for others. In this study, we measure ED crowding by wait time and address two key questions: (1) What is the average effect of Medicaid expansion on ED wait time? (2) How does the effect of Medicaid expansion vary across different hospitals and populations? The first question is of interest to policymakers, especially those in nonexpansion states, as they evaluate the overall benefit of Medicaid expansion. The second question is of interest to hospital managers because they want to know the effect of Medicaid expansion on their hospitals.

We contribute to existing literature on Medicaid expansion and ED operations management by empirically investigating the effect of Medicaid expansion on ED wait time. We use nonexpansion states as a control group and compare ED wait times in expansion states before and after the expansion. We find that Medicaid expansion increases ED wait time by 10.4% (or 3.5 minutes). To analyze the heterogeneous effects of Medicaid expansion, we first develop a novel first-difference causal forest (FDCF) approach that incorporates the first-difference approach into the causal forest approach. We then apply the FDCF approach to our empirical example and find that the effect of Medicaid expansion varies widely across different hospitals.

To the best of our knowledge, this paper is the first to study the effect of Medicaid expansion on ED wait time and find that the effect is heterogeneous across different hospitals. The results are useful to policymakers seeking to understand the effect of Medicaid expansion on ED wait time and identify potential areas for improvement. The results are also useful to hospital managers in helping them understand how Medicaid expansion affects their hospitals. Finally, our results are useful to researchers analyzing heterogeneous effects of a treatment when the number of features is large.

2. Literature Review

Our study is related to two streams of literature: (1) the effects of Medicaid expansion; and (2) ED operations management. The first stream of literature looks at Medicaid expansion from a health policy perspective and studies the effect of Medicaid expansion on insurance coverage, care utilization, and wait time. The second stream of literature looks at ED patient flow from an operations management perspective and studies how to reduce wait time, treatment time, and length of stay. We briefly review these two streams of literature and highlight our contributions.

2.1. Effects of Medicaid Expansion

A large body of health economics and medical literature investigates the impact of Medicaid expansion on access to care, affordability, and various economic measures (see Antonisse et al. 2018 and Mazurenko et al. 2018 for a comprehensive literature review). In this section, we focus on literature that relates to ED utilization, including payer mix and total number of patient visits. We also review literature that studies the impact of Medicaid expansion on the availability of primary care physicians and wait time to appointment.

Interest in understanding whether Medicaid expansion has changed the mix of patients at hospitals has been growing. In particular, many studies focus on the insurance type of the patients admitted to EDs (Pines et al. 2016, Barakat et al. 2017, Dresden et al. 2017, Klein et al. 2017, Nikpay et al. 2017, Sabik et al. 2017, Ladhania et al. 2021). Almost all of these studies find that Medicaid expansion increases the number of Medicaid ED visits and reduces the number of uninsured ED visits. The results of this line of literature suggest that Medicaid expansion has helped uninsured patients gain insurance coverage—an important goal of the ACA.

Although most research demonstrates that Medicaid expansion has increased insurance coverage, findings on the total number of ED visits are mixed. For example, Dresden et al. (2017) and Nikpay et al. (2017) find that Medicaid expansion increases the total number of ED visits. By contrast, Klein et al. (2017) find that Medicaid expansion decreases the total number of ED visits. In two other studies, Pines et al. (2016) and Sabik et al. (2017) find no significant relationship between Medicaid expansion and the total number of ED visits.

These discrepancies between studies could be due to three reasons. First, these studies use data from different states. The effect of Medicaid expansion may be positive for some states, but negative or insignificant for others. Second, these studies focus on different time frames. The effect of Medicaid expansion may be small in the short term, but large in the long term. Third, some of these studies compare the total number of ED visits in the same state before and after Medicaid expansion. This approach could lead to biased estimates, because the results include changes due to time trends that are present, even in the absence of the expansion.

Similarly, findings about the effect of Medicaid expansion on physician availability and wait time are mixed. For example, Greener et al. (2019) and Miller and Wherry (2017) find that Medicaid expansion reduces physician availability and increases wait time, whereas Carey et al. (2020), Polsky et al. (2015), and Tipirneni et al. (2016) find that physician availability and wait time remain almost the same. A few other studies perform subgroup analyses and find that Medicaid expansion increases wait time for patients with private insurance, but reduces the wait time for patients with Medicaid insurance (see, e.g., Tipirneni et al. 2016 and Anandasivam et al. 2017).

Surprisingly, none of the existing studies examine the effect of Medicaid expansion on ED wait time, which is an important quality metric that affects both patient experience and medical outcomes. This lack in the literature is partly due to unavailability of data, because many hospitals do not track the wait time of individual patients in their EDs. However, with the recent release of the Hospital Compare data by the Centers for Medicare and Medicaid Services (CMS), we are able to bridge this gap by examining the effect of Medicaid expansion on ED wait time.

2.2. ED Operations Management

Existing studies in the operations management literature find that patients are sensitive to ED waiting and take wait time into account when choosing an emergency service provider (see, e.g., Dong et al. 2019). Prolonged wait time remains a concern, even after patients are admitted to an ED. Patients are more likely to abandon a queue when they observe more patients in it, more patients arriving, and fewer patients leaving the ED (Batt and Terwiesch 2015). Besides influencing the choice and behavior of patients, wait time affects the decisions of care providers. For example, studies find that ED decision makers take wait time into consideration when classifying patients into different triage levels (see, e.g., Ding et al. 2019).

The wait time in an ED is difficult to predict, because the demand for emergency service is uncertain. Also, because patients have different levels of sickness and physicians have different levels of efficiency, the time a physician takes to assess and treat a patient can be uncertain as well. To address these challenges and better predict wait time, Ang et al. (2015) develop a new approach that combines queueing theory with the least absolute shrinkage and selection operator method to predict ED wait time. The authors show that their approach has a better prediction than existing approaches, such as rolling-average methods, fluid model estimators, and quantile regression methods.

Instead of predicting wait time, some studies examine various provider practices that could reduce ED wait time or length of stay and improve medical outcomes associated with these practices. For example, Song et al. (2015) compare dedicated and pooled queueing systems and find that a dedicated queue increases the ownership of physicians over patients and reduces wait time and length of stay. Batt and Terwiesch (2016) find that early task initiation (e.g., having tests ordered at the triage process) reduces treatment time, but increases the total number of tests performed on the patient. KC (2013) finds that physician multitasking first reduces and then increases (i.e., follows a U-shape) the total time taken to discharge a given number of patients, and excessive multitasking results in a smaller number of detected diagnoses and a higher rate of readmission.

Interest in applying analytical models to reduce wait time or balance workload in an ED has been growing. This literature looks at waiting as a scheduling problem and uses queueing or optimization models to find ways to reduce ED crowding. For example, Xu and Chan (2016) propose a prediction-guided diversion policy that first predicts potential patient arrivals in the future and then utilizes the information to proactively divert patients before an ED becomes congested. He et al. (2019) propose a hybrid robust stochastic approach to maximize the percentage of patients whose wait time and length of stay are within the mandatory targets. Mandelbaum et al. (2012) propose a routing policy that requires only limited and partial information to achieve a long-run workload balance across medical staff.

Other studies examine how triage and patient streaming can help improve patient flow in the ED. For example, Kamali et al. (2018) use a queueing model to analyze the impact of physician versus nurse triage on patient flow and find that nurse triage outperforms physician triage when patient arrival rate is either very low or very high. Saghafian et al. (2012) analyze the impact of patient streaming on the responsiveness and safety of an ED and find that streaming improves patient flow when ED resources are shared across streams and for EDs with a high percentage of admitted patients, long patient boarding time, high average physician utilization, high day-to-day variation in the percentage of admitted patients, and longer stays for admitted patients than discharged patients.

3. Background, Clinical Setting, and Data

We first review the background of the ACA, which includes a provision to expand Medicaid. We then describe patient flow in the ED and related metrics to measure ED crowding. Then, we discuss the potential impact of Medicaid expansion on ED wait time. Finally, we describe the data sets used in this study and discuss how we prepare them for empirical analyses.

3.1. Affordable Care Act and Medicaid Expansion

The ACA is a healthcare reform law that President Obama signed into law in 2010 in response to concerns about the growing number of uninsured patients and rising healthcare costs. It contains 10 titles to address a number of issues, including the cost and quality of care, public health and prevention of disease, and new taxes to pay for the expansion of insurance coverage.

The ACA has three primary goals: (1) lower healthcare costs through innovative medical delivery methods and better healthcare decision making; (2) reform the private insurance market and provide more affordable health insurance to more people; and (3) expand Medicaid to cover all adults with incomes below 138% of the federal poverty level (Kaiser Family Foundation 2013a).

Among the three goals, Medicaid expansion is the most controversial, because the federal government pays 100% for newly eligible enrollees for the first three years and gradually reduces its share to 90% by 2020. Therefore, individual states will need to bear as much as 10% of the cost associated with Medicaid expansion—a significant expenditure that some states could not afford.

When the ACA was signed into law in 2010, it mandated that all states expand Medicaid to their population. The federal government would reduce existing Medicaid funding if a state did not participate in the expansion. However, in 2012, the Supreme Court ruled that states could not be forced to participate in a new healthcare program, which made Medicaid expansion an optional program.

3.2. ED Patient Flow and Outcome Measures

A typical patient flow in the ED is depicted in Figure 1. It starts when a patient arrives at an ED from ambulatory care or via emergency medical services. After registration and triage, the patient sees a healthcare professional (usually a nurse practitioner or physician assistant), who takes vital signs, orders diagnostic tests (e.g., blood tests or X-ray), and assigns a medical doctor based on the patient's condition and the doctor's specialty. The patient then sees the assigned doctor, who assesses the patient's condition, provides medical treatment, and decides whether the patient should be admitted to an inpatient department or discharged to go home. If the doctor decides the patient should be admitted as an inpatient, the patient needs to stay in the ED until an inpatient bed becomes available.

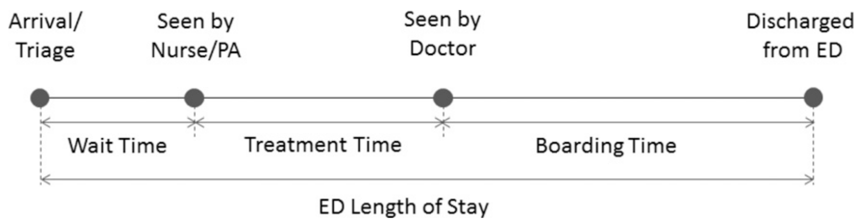
ED crowding can be measured in four main ways. The first measure is ED wait time, which is defined as the time between patient arrival and the patient seeing a healthcare professional. The second measure is ED treatment time, which is defined as the time between a patient seeing a nurse or physician assistant and the patient finishing seeing a doctor. The third measure is boarding time, which is defined as the time between a doctor deciding to admit a patient to an inpatient department and the patient being admitted. The final measure is ED length of stay, which is defined as the time between patient arrival and the patient being released from the ED. In this study, we focus on ED wait time in the main analysis. As a robustness check, we use boarding time and ED length of stay as alternative outcome measures.

3.3. Impact of Medicaid Expansion on ED Wait Time

Existing studies (see, e.g., Green et al. 2006 and Chowdhury et al. 2018) analyze ED wait time using queueing models. The findings from these queueing models suggest the following: (1) An increase in the number of ED visits increases ED wait time; (2) an increase in the average treatment time increases ED wait time; and (3) an increase in ED utilization, which is a function of the number of ED visits and average treatment time, exponentially increases ED wait time. In this section, we first discuss how Medicaid expansion affects the number of ED visits and average treatment time. We then discuss why the effect may be heterogeneous across hospitals.

Medicaid expansion can increase ED wait time through two channels. First, the expansion can increase the demand for care and, therefore, the number of ED visits. Before the expansion, uninsured patients had a low demand for healthcare resources. Some of these patients avoided visiting healthcare providers because they could not afford high medical expenses (Buchmueller et al. 2005). Others did not know where

Figure 1. Patient Flow in the Emergency Department



Notes. This figure depicts a typical patient flow in the ED. Wait time is defined as the time between patient arrival and the patient seeing a healthcare professional. Treatment time is defined as the time between a patient seeing a nurse or physician assistant (PA) and the patient finishing seeing a doctor. Boarding time is defined as the time between a doctor deciding to admit a patient to an inpatient department and the patient being admitted. ED length of stay is defined as the time between patient arrival and the patient being released from the ED.

and how to seek care, due to a lack of medical literacy or transportation (Sentell 2012). Because the expansion improved the access to care and lowered the cost to patients, existing studies find that patients tend to use more healthcare resources after they obtain health insurance (Miller 2012, Baicker et al. 2013). In particular, the randomized controlled Oregon Health Insurance Experiment provides compelling evidence that Medicaid coverage increases the number of ED visits across a broad range of visit types, conditions, and populations (Finkelstein et al. 2012, Taubman et al. 2014).

Second, the expansion can increase the proportion of sicker patients and, therefore, the average treatment time. Recall that treatment time is defined as the time between a patient seeing a nurse or physician assistant and the patient finishing seeing a doctor. Given the same healthcare providers, the treatment time for sicker patients is longer than for healthier patients because sicker patients have more complicated conditions that require a larger number of diagnoses and tests. Uninsured patients have poorer health conditions than insured patients, partly because the former have limited access to healthcare resources, due to a lack of health insurance before the expansion (see, e.g., McWilliams 2009). After the expansion, these patients are more likely to use healthcare resources, including EDs. As a result, the proportion of sicker patients, and, therefore, the average treatment time, increases.

Our discussion so far suggests that Medicaid expansion increases ED wait time. However, the expansion can reduce ED wait time by shifting the demand from EDs to primary care physicians. Prior to Medicaid expansion, some uninsured patients relied on emergency service as their primary healthcare resource, because the Emergency Medical Treatment and Active Labor Act prohibits an ED from denying a patient because of her insurance (Zibulewsky 2001). Because a sizeable number of uninsured patients visit EDs even when they do not have emergent issues, some studies cite inappropriate use of EDs as an important cause of ED crowding (Morley et al. 2018). With Medicaid expansion, these uninsured patients would switch from emergency care to other types of care (e.g., primary care), which reduces the number of ED visits and, thus, ED wait time (Burke and Paradise 2015). Therefore, whether the expansion reduces or increases ED wait time is ultimately an empirical question.

If Medicaid expansion increases ED wait time through a change in the number of ED visits and average treatment time, we expect the effect to be heterogeneous across hospitals, depending on the characteristics of both demand and supply sides. On the demand side, the number of patients who qualify for Medicaid expansion is determined by their incomes relative to the federal poverty level, which suggests that the increase in the number of Medicaid patients will be larger in poor areas than in rich areas (Garfield et al. 2021). The proportion of these newly enrolled Medicaid patients using emergency care will also depend on patients' health and ease of access to care. For example, older and sicker patients with cars may be more willing to go to EDs than younger and healthier patients who rely on public transportation (Sieck et al. 2016).

On the supply side, hospitals have different characteristics, such as teaching status, insurance policies, and information technology systems. Hospitals (e.g., those with teaching status) that are more attractive to patients are more likely to see an increase in the number of patients (Ayanian and Weissman 2002). However, whether a given hospital will see an increase in the number of patients in the ED or other departments depends on the hospital's willingness to admit Medicaid patients to its nonemergency departments. Finally, hospitals have different capabilities for reducing crowding in the face of increased demand. For example, larger hospitals with an electronic health record system may be better able to manage crowding than smaller hospitals without such an asset (Definitive Healthcare 2019).

3.4. Data Description and Preparation

Our primary data set is obtained from the Hospital Compare website, which compares the quality of care at Medicare-certified hospitals across the country based on the data submitted by hospitals to the CMS Clinical Data Warehouse through the CMS Abstraction and Reporting Tool.[1] The quality metrics used by Hospital Compare include patients' experiences, timely and effective care, complications, readmissions and deaths, use of medical imaging, and payment and value of care. Hospital Compare allows a patient to compare two or more hospitals based on the quality metric selected by the patient.

The timely and effective care metric measures the proportion of patients who get appropriate treatments for certain common, serious medical conditions or surgical procedures; how quickly hospitals provide treatments to patients with certain medical emergencies; and how well hospitals provide preventive services. It examines five types of care—emergency care, influenza vaccination, blood clot, pregnancy, and delivery. This study focuses on emergency care. Note that the ED data collected by CMS are based on the emergency care received by both Medicare and non-Medicare patients.

One useful feature of this data set is that it includes detailed hospital-level outcome information (e.g., ED wait time, boarding time, and length of stay), which allows us to compare the outcomes of different hospitals. Another helpful feature is that the data set is updated quarterly, so we can compare the periods before and after Medicaid expansion.[2] Finally, this data set includes states that expanded Medicaid and those that did not, which allows us to study the effect of Medicaid expansion using nonexpansion states as a control group.

Although the Hospital Compare data set provides rich details about the outcomes of different hospitals, it does not include information about the characteristics of patients (e.g., demographics, education, and sickness), which may be important to control for when we examine the outcome of an ED. To address this issue, we supplement the Hospital Compare data set with the data set from the American Community Survey (ACS), which gathers zip-code-level information on a yearly basis about population demographics; jobs and occupations; educational attainment; whether people own or rent their homes; and other topics.[3] We use these zip-code-level population characteristics as proxies for the characteristics of patients at a hospital with the same zip code.

Because we are interested in understanding how different types of hospitals are affected differently by Medicaid expansion, we supplement the Hospital Compare data set further with the data set from the American Hospital Association (AHA).[4] The AHA data set is an annual survey of 6,500 hospitals in the United States. It provides detailed information about organizational structure, personnel, hospital facilities and services, utilization data, physician arrangements, staffing, and community orientation.

The Hospital Compare data set includes 2,709 hospitals with outcome information across years between 2012 and 2016. We link the data set from Hospital Compare with those from ACS and AHA using hospital identifiers and zip codes provided by these data sets. We exclude 191 hospitals that cannot be matched among the three data sets.[5] Because we focus on the states that either expanded Medicaid at the beginning of 2014 or did not expand Medicaid before 2016 in the main analysis, we exclude an additional 282 hospitals in the states that expanded Medicaid between 2014 and 2016. Our final sample includes 11,180 observations (or 2,236 hospitals across five years).

4. Econometric Model

To study the effect of Medicaid expansion on ED wait time, we note that the decision of whether to expand Medicaid may be determined endogenously. For example, states with more insured patients and better fiscal stability may be more likely to expand Medicaid. At the same time, these states may have a shorter ED wait time. Therefore, a direct comparison of the states that expanded Medicaid and those that did not may be subject to unobservable confounding factors that affect both Medicaid expansion and ED wait time. An alternative approach is to compare pre-expansion and postexpansion ED wait times of the states that expanded Medicaid. But this approach leads to biased estimates as well, because the results include changes due to time trends that are present even in the absence of the expansion.

To isolate some of these confounding factors, our empirical strategy relies on identifying the states that did not expand Medicaid and using them as a control group. The main idea behind this strategy is that these nonexpansion states experienced similar time trends, but were not subject to the treatment of Medicaid expansion. We can then compare the pre-expansion and postexpansion periods for both the treatment and control groups, using the difference-in-differences approach, to estimate the effect of Medicaid expansion. More specifically, our treatment group includes 25 states that expanded Medicaid at the beginning of 2014, and our control group includes 20 states that did not expand the Medicaid program during our study period (i.e., from 2012 to 2016).[6]

A critical assumption of the difference-in-differences approach is that the treatment and control groups have parallel trends in the absence of the treatment. This assumption may be violated if unobservable features that affect both Medicaid expansion and ED wait time exist. We address this potential concern in four steps. First, we include a large number of hospital characteristics (e.g., teaching status, rural/urban, and electronic health record implementation) and population characteristics (e.g., demographics, education, and employment) as independent variables. Second, we perform extensive robustness checks by including features such as health exchanges, states' political party affiliation, and the woodwork effect as additional independent variables. Third, we follow existing literature (see, e.g., Brot-Goldberg et al. 2017, Bapna et al. 2018, and Sun et al. 2020) to address potential endogeneity issues by using early adopters as a control group and late adopters as a treatment group. Finally, we follow Rosenbaum and Silber (2009) by performing sensitivity analyses to determine how strong the effects of the unobservable features would have to be to change the substantive conclusion from our study.

4.1. Average Effect Analysis

To describe the difference-in-differences approach, we index hospital by i and year by t. Denote by $Hospital_i$ and $Year_t$ vectors of hospital and year dummies, respectively. Denote by $Medicaid_{it}$ a binary variable that equals one if hospital i is in a state that expanded Medicaid in year t or earlier and zero otherwise. Denote by X_{it} the characteristics of hospital i in year t. We can describe the outcome variable Y_{it} (e.g., ED wait time) as

$$Y_{it} = \alpha + \beta Medicaid_{it} + \gamma X_{it} + \sigma Hospital_i + \lambda Year_t + \varepsilon_{it}, \tag{1}$$

where ε_{it} denotes an idiosyncratic error.

The coefficient of $Medicaid_{it}$ (i.e., β) captures the causal effect of Medicaid expansion on ED wait time. A positive coefficient would indicate that Medicaid expansion increases ED wait time, whereas a negative coefficient would indicate otherwise. Note that we include hospital fixed effects to capture systematic differences across different hospitals and year fixed effects to control for time trends across different years.

To address potential concerns that hospital or population characteristics may change due to Medicaid expansion, we consider four different models, in which X_{it} includes (1) no hospital or population characteristics, (2) only hospital characteristics, (3) only population characteristics, and (4) both hospital and population characteristics. Comparing the results from these models allows us to understand the sensitivity of our results.

4.2. Heterogeneous Effect Analysis

The difference-in-differences regression model described above does not capture potential treatment-effect heterogeneity across different hospitals. Medicaid expansion may have a positive effect on some hospitals, but a negative or no effect on others. To analyze the heterogeneous effects of Medicaid expansion, we develop an FDCF approach by incorporating the first-difference approach into the casual forest approach. In this section, we first describe the standard causal forest approach. We then discuss the performance of the proposed FDCF approach and how to apply it to our empirical example.

4.2.1. Causal Forest Approach. To describe the causal forest approach, we denote by X_i the characteristics of hospital i and by W_i a treatment dummy that equals one if hospital i is in an expansion state and zero otherwise. Following the potential outcomes framework (see, e.g., Rubin 1974 and Abadie 2005), we denote by $Y_i^{(1)}$ and $Y_i^{(0)}$ the potential outcomes that hospital i would have experienced with and without the treatment, respectively. We define the effect of Medicaid expansion for hospital i as $\beta_i = Y_i^{(1)} - Y_i^{(0)}$. We have $\beta_i \neq \beta_j, \exists i \neq j$ if the effect is heterogeneous, and $\beta_i = \beta_j, \forall i \neq j$ if the effect is homogeneous. The fundamental problem of causal inference is that we do not observe both potential outcomes $Y_i^{(1)}$ and $Y_i^{(0)}$ for a given hospital. Instead, we observe only the realized outcome $Y_i = W_i Y_i^{(1)} + (1 - W_i) Y_i^{(0)}$, so we cannot compute the individual-level treatment effect β_i.

A common approach to addressing this fundamental problem is to temporarily ignore individual-level treatment effects and focus on conditional average treatment effects (see, e.g., Athey et al. 2019). For a given value of hospital characteristics x, the objective is to estimate the treatment effect $\beta(x) = E[Y_i^{(1)} - Y_i^{(0)} \mid X_i = x]$. Achieving this objective presents two major challenges: (1) A forest might focus its splits on the features that affect the main outcome (i.e., main effects) instead of the features that affect the treatment effect (i.e., treatment-effect heterogeneity); and (2) how to identify an appropriate set of neighbors and use them to estimate the conditional average treatment effect $\beta(x)$ is unclear.

To overcome the first challenge, Athey et al. (2019) use the local centering approach to regress out the effect of the characteristics X_i on the outcome Y_i and treatment W_i separately. Denote by $y(x) = E[Y_i \mid X_i = x]$ and $w(x) = E[W_i \mid X_i = x]$ the marginal expectations of the outcome and treatment, respectively. The authors first leave out the i-th observation and calculate the leave-one-out estimates of the marginal expectations $\hat{y}^{-i}(X_i)$ and $\hat{w}^{-i}(X_i)$ and then calculate the centered outcome $\widetilde{Y}_i = Y_i - \hat{y}^{-i}(X_i)$ and treatment $\widetilde{W}_i = W_i - \hat{w}^{-i}(X_i)$. The authors show that using the centered, instead of the original, outcome and treatment allows a forest to focus its splits on the features that affect the treatment effect, instead of the main outcome.

To overcome the second challenge, the authors first develop a forest-based approach to calculate a similarity weight and then apply the local generalized method of moments to estimate $\beta(x)$ based on a weighted set of neighbors. For purposes of illustration, we describe first the local generalized method of moments and then the forest-based approach. Denote by $\alpha_i(x)$ a similarity weight that measures the relevance of the i-th observation to x. We can estimate $\beta(x)$ by solving the following equation:

$$\hat{\beta}(x) = \left[\sum_{i=1}^{n} \alpha_i(x)\left(\widetilde{W}_i - \overline{W}_\alpha\right)\left(\widetilde{W}_i - \overline{W}_\alpha\right)^T\right]^{-1} \sum_{i=1}^{n} \alpha_i(x)\left(\widetilde{W}_i - \overline{W}_\alpha\right)\left(\widetilde{Y}_i - \overline{Y}_\alpha\right), \tag{2}$$

where $\overline{W}_\alpha = \sum \alpha_i(x)\widetilde{W}_i$ and $\overline{Y}_\alpha = \sum \alpha_i(x)\widetilde{Y}_i$.

A traditional approach to calculating $\alpha_i(x)$ is based on some kernel function that places a larger weight on nearby observations. This approach is prone to a curse of dimensionality when the number of characteristics is large. The causal forest addresses this challenge by replacing the kernel weighting with a forest-based approach. It calculates the similarity weight based on the frequency with which the i-th observation falls into the same leaf as x in causal trees. Denote by B the number of trees in a causal forest and by $L_b(x)$ the set of observations falling in the same leaf as x in the b-th tree. The similarity weight α_i can be calculated as

$$\alpha_i(x) = \frac{1}{B}\sum_{b=1}^{B}\frac{\mathbb{1}[X_i \in L_b(x)]}{|L_b(x)|}. \tag{3}$$

We now discuss how the causal forest approach constructs individual causal trees. Denote by n_P the number of observations in parent node P and by n_{C_1} and n_{C_2} the number of observations in candidate child nodes C_1 and C_2. Denote by $\hat{\beta}_{C_1}$ and $\hat{\beta}_{C_2}$ the treatment effects for the two child nodes estimated from Equation (2). A causal tree partitions observations in a way that increases the heterogeneity of in-sample β-estimates as fast as possible based on the criterion $\Delta(C_1, C_2) := n_{C_1}n_{C_2}/n_P^2(\hat{\beta}_{C_1} - \hat{\beta}_{C_2})^2$. Because optimizing the criterion over all possible ways of partitioning while explicitly solving for $\hat{\beta}_{C_1}$ and $\hat{\beta}_{C_2}$ in individual child nodes is computationally expensive, the causal forest approach uses a gradient-based approximation method that calculates a pseudo-outcome using

$$\rho_i = \left[\frac{1}{|\{i : X_i \in P\}|}\sum_{\{i:X_i\in P\}}\left(\widetilde{W}_i - \overline{W}_P\right)\left(\widetilde{W}_i - \overline{W}_P\right)^T\right]^{-1} \left(\widetilde{W}_i - \overline{W}_P\right)\left(\widetilde{Y}_i - \overline{Y}_P - \left(\widetilde{W}_i - \overline{W}_P\right)\hat{\beta}_P\right), \tag{4}$$

where $\hat{\beta}_P$ is estimated from Equation (2), and $\overline{W}_P$ and $\overline{Y}_P$ are the averages taken over the parent node P.

It then treats the pseudo-outcome ρ_i as a new outcome variable and uses the standard classification and regression-tree method to partition observations in a parent node into two child nodes (Breiman et al. 1984). Then, the causal tree treats each of the two child nodes as a parent node and repeats the process of constructing pseudo-outcomes and partitioning nodes until a stopping criterion is reached. Athey et al. (2019) show that estimates from the causal forest approach are consistent and asymptotically Gaussian and provide an estimator for their asymptotic variance that allows us to construct valid confidence intervals.

To illustrate the application of the causal forest approach, we consider two hospital groups (i.e., expansion and nonexpansion) across two time periods (i.e., pre-expansion and postexpansion) and divide observations into four categories: (1) nonexpansion group in pre-expansion period, (2) nonexpansion group in postexpansion period, (3) expansion group in pre-expansion period, and (4) expansion group in postexpansion period. We denote the outcome variables of these four categories of observations by $Y_{i0}^{(0)}$, $Y_{i1}^{(0)}$, $Y_{j0}^{(0)}$, and $Y_{j1}^{(1)}$, respectively, where subscripts i and j are indexes of hospitals in the nonexpansion and expansion groups, respectively, and subscripts 0 and 1 indicate whether the outcome is measured during the pre-expansion or postexpansion period.

Because the causal forest approach was designed to analyze heterogeneous effects of a binary treatment, one may be tempted to compare two categories of observations each time. One approach (designated as "CF1") is to compare ED wait times of hospitals in nonexpansion and expansion groups during the postexpansion period (i.e., the difference between categories (2) and (4))—that is, $\widetilde{\beta}(x) = E[Y_{j1}^{(1)} - Y_{i1}^{(0)} \mid X_{i1} = X_{j1} = x]$. Another approach (designated as "CF2") is to compare ED wait times of hospitals in the expansion group before and after the expansion (i.e., the difference between categories (3) and (4))—that is, $\check{\beta}(x) = E[Y_{j1}^{(1)} - Y_{j0}^{(0)} \mid X_{j0} = X_{j1} = x]$. Both approaches may lead to biased estimates because the conditional average treatment effect should be expressed as $E[\beta(x)] = \{E[Y_{j1}^{(1)} \mid X_{j1} = x] - E[Y_{j0}^{(0)} \mid X_{j0} = x]\} - \{E[Y_{i1}^{(0)} \mid X_{i1} = x] - E[Y_{i0}^{(0)} \mid X_{i0} = x]\}$ (see, e.g., Heckman et al. 1997 and Abadie 2005). This difference-in-differences estimator motivates our FDCF approach, which we discuss next.

4.2.2. FDCF Approach. We develop a two-step FDCF approach by incorporating the first-difference approach into the causal forest approach. In the first step, we use the first-difference approach to calculate the change in ED wait time of each hospital from the pre-expansion to postexpansion period—that is, we subtract category (1) from category (2) to obtain $Y_{i1}^{(0)} - Y_{i0}^{(0)}$ (denoted as Y_i) and subtract category (3) from category (4) to obtain $Y_{i1}^{(1)} - Y_{j0}^{(0)}$ (denoted as Y_j). This step teases out the systematic differences between hospitals in nonexpansion and expansion groups. In the second step, we use the outcome variables (i.e., Y_i and Y_j) obtained from the first step and apply the causal forest approach for heterogeneous treatment-effect analysis. This step teases out the effect of time trends that exist in both the treatment and control groups.

We conduct comprehensive simulation studies to assess the performance of FDCF, as well as CF1 and CF2 approaches. We are interested in understanding whether the CF1 and CF2 approaches lead to biased estimates and, if so, whether the FDCF approach is effective in correcting the bias. A major advantage of simulation studies is that we know what the true treatment effects are, so we can compare estimated with true treatment effects to see how much they differ from each other. Following existing machine literature (see, e.g., Athey et al. 2019), we use mean-squared error (MSE) to evaluate the performance of different approaches. Mathematically, denote by N the sample size, and by $\hat{\beta}_i$ and β_i the estimated and true treatment effects, respectively. We calculate the MSE as $MSE = \frac{1}{N}\sum_{i=1}^{N}(\hat{\beta}_i - \beta_i)^2$. A model has a better performance if it has a smaller MSE.

We consider 12 designs (see Table B1 in Online Appendix B) to analyze how the performance of FDCF, CF1, and CF2 approaches changes with (1) feature type (compare designs 1 and 2), (2) error distribution (compare designs 1, 3, and 4), (3) correlation of features (compare designs 1 and 5), (4) model specification (compare designs 6–10), (5) homogeneous treatment effect (compare designs 1 and 11), and (6) impact of features (compare designs 1 and 12). Each design has a total of 10 features, where x_{ikt} and z_{il} denote the features that affect the main outcome and treatment effect, respectively. Features that do not affect the mean outcome or treatment effect are excluded from the model, but are included in the data for heterogeneous treatment-effect analysis. To understand the impact of sample size, we increase the number of observations incrementally from 1,000 to 5,000.

The results from the simulations studies are summarized in Table B2 in Online Appendix B. They are calculated based on a testing sample size of 5,000 and 10 rounds of the simulations. The number of trees in a forest is 4,000, and the minimum node size is 50 for all approaches. From the table, we see that the performance of the FDCF approach is superior to that of the CF1 and CF2 approaches. For example, in design 1 with a sample size of 5,000, the relative gap between the FDCF and CF1 approaches is $(1.22 - 0.22)/1.22 = 82\%$ and between the FDCF and CF2 approaches is $(5.69 - 0.22)/5.69 = 96\%$. The performance gap is larger when features are binary instead of continuous, due to the natural subgroups implied by the binary features (see design 2). By comparing designs 1 and 4–6, we see that the superior performance of the FDCF approach is robust to different error distributions and correlations between features. Surprisingly, our model performs relatively well, even when the underlying models are nonlinear (see designs 7–10). Finally, the FDCF approach has an even better performance when the treatment effect is homogeneous or no feature affects mean outcomes (compare designs 11 and 12 with design 1).

To apply the FDCF approach to our empirical example, we consider a broad set of hospital and population characteristics for heterogeneous treatment-effect analysis. The hospital characteristics include teaching status, rural/urban, electronic health record (EHR) implementation, free-standing ED, health maintenance organization (HMO)/preferred provider organization (PPO), number of beds, and number of physicians. The population characteristics include age, gender, race, education, labor percentage, unemployment rate, insurance, household income, and population size. Note that some of these characteristics remain the same during our study period, whereas others change over time. Because the change in characteristics may be due to Medicaid expansion, we use pre-expansion characteristics in the FDCF approach for heterogeneous treatment-effect analysis.

5. Results and Discussion

We first provide summary statistics to compare expansion and nonexpansion states in the periods before and after Medicaid expansion. We then describe the results from average effect analysis. Finally, we describe the results from heterogeneous effect analysis and show that the effect of Medicaid expansion is heterogeneous across different hospitals.

5.1. Descriptive Results

Thirty-one states (27 in 2014, three in 2015, and one in 2016) expanded Medicaid during our study period (see Table C1 in Online Appendix C for more details). Among the states that expanded Medicaid in 2014, Michigan and New Hampshire expanded Medicaid in April and August, respectively. In the main analysis, we use the other 25 states, which expanded Medicaid at the beginning of 2014, collectively as a treatment group. The control group includes the 20 states that did not expand Medicaid during our study period. As a robustness check, we include all states and re-estimate the effect of Medicaid expansion.

We compare average ED wait times of the treatment and control groups before and after Medicaid expansion.[7] The upper left side of Table 1 summarizes the unweighted results in 2013 and 2015. We see a decrease in ED wait time in both treatment and control groups (as indicated by the column of "Difference"). By comparing the changes in the treatment and control groups, we see that the treatment group has a smaller decrease in ED wait time than the control group (see the row of "Difference"). These results suggest that Medicaid expansion increases ED wait time. We observe similar changes when using population-weighted results (see the lower panel of Table 1) or comparing ED wait times two years before and after the expansion (see the right side of Table 1).

5.2. Average Treatment Effects

Whereas the descriptive results presented in Section 5.1 suggest that Medicaid expansion increases ED wait time, they do not properly control for the differences in hospital and population characteristics. In this section, we formally study the effect of Medicaid expansion using the difference-in-differences approach described in Section 4.1. To allow for differences in the variance/standard errors due to arbitrary intragroup correlation, we have robust standard errors clustered by state (KC and Terwiesch 2011, Berry Jaeker and Tucker 2016).

We use the log-transformed ED wait time as the dependent variable of the regression model (see Equation (1) in Section 4.1) because the distribution of ED

Table 1. Comparison of ED Wait Times Before and After Medicaid Expansion

	One year before and after			Two years before and after		
Group	Before	After	Difference	Before	After	Difference
Unweighted						
Expansion	31.77	29.42	−2.35	32.99	28.35	−4.64
Nonexpansion	30.79	25.69	−5.10	32.03	24.67	−7.36
Difference	0.98	3.73	2.75	0.96	3.68	2.72
Population weighted						
Expansion	33.85	31.60	−2.25	35.21	30.47	−4.74
Nonexpansion	30.83	25.55	−5.28	32.23	24.54	−7.69
Difference	3.02	6.05	3.03	2.98	5.93	2.95

Notes. This table compares average ED wait times of the treatment and control groups before and after Medicaid expansion based on the 2012–2016 data from Hospital Compare. The upper and lower panels of the table summarize the unweighted and population-weighted results, respectively. The left side of the table compares the results in 2013 and 2015, and the right side of the table compares the results in 2012/2013 and 2015/2016.

wait time is right-skewed. From Table 2, we see that the coefficients of *Medicaid Expansion* are positive and statistically significantly different from zero at the 1% significance level, which suggests that Medicaid expansion increases ED wait time. Because we use the log-transformed ED wait time as the dependent variable, these coefficients can be interpreted as percentage changes in ED wait time due to Medicaid expansion. For example, in model (4), a coefficient of 0.104 can be interpreted as a 10.4% increase in ED wait time. The results from models (1)–(4) are not statistically significantly different from each other at the 10% significance level, which addresses potential concerns that hospital or population characteristics may change due to Medicaid expansion.

5.3. Robustness Check

We perform extensive robustness checks to (1) formally test the parallel-trends assumption made in the difference-in-differences model; (2) understand how the effect of Medicaid expansion changes over time; (3) check whether propensity score matching improves our results; (4) address potential endogeneity issues caused by unobservable hospital or population characteristics; (5) address a potential concern that our data set has few time periods and many subjects; (6) test whether our results are driven by a particular state or a group of states; (7) disentangle the effect of Medicaid expansion from that of health exchanges; (8) address potential concerns related to states' political party affiliation; and (9) analyze the effect of Medicaid expansion on boarding time and ED length of stay. All results are provided in Online Appendix E.

Table 2. Average Effect of Medicaid Expansion on ED Wait Time (Log-Transformed)

	Model			
Variable	(1)	(2)	(3)	(4)
Medicaid expansion	0.106***	0.105***	0.106***	0.104***
	(0.038)	(0.039)	(0.038)	(0.039)
Hospital characteristics	No	Yes	No	Yes
Population characteristics	No	No	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Hospital fixed effects	Yes	Yes	Yes	Yes
Number of observations	11,180	11,180	11,180	11,180
Adjusted R^2	0.745	0.745	0.745	0.746

Notes. This table summarizes the results based on the 2012–2016 data from Hospital Compare, AHA, and ACS. The number of observations refers to the number of unique hospital-year pairs. Model (1) does not include hospital or population characteristics. Models (2) and (3) include hospital and population characteristics, respectively. Model (4) includes both hospital and patient characteristics. Standard errors are clustered by state.

***$p \leq 0.01$.

5.3.1. Parallel-Trends Assumption.

To formally test the parallel-trends assumption, we modify the difference-in-differences model by including a vector of pretreatment dummies. More specifically, we denote by $Medicaid_{it}^{-j}$, where $j = \{1, 2\}$, a binary variable that equals one if the state of hospital i expanded Medicaid j years after year t and zero otherwise. We use the dummy $Medicaid_{it}^{-2}$ as a control and test whether the coefficient of $Medicaid_{it}^{-1}$ is statistically significantly different from zero. A significant coefficient would suggest the treatment and control groups do not have parallel trends before Medicaid expansion, whereas an insignificant coefficient would indicate otherwise.[8]

The results from the modified model are summarized in Table E1 in the online appendix. We see that the coefficients of the pretreatment dummy $Medicaid_{it}^{-1}$ have small magnitudes and are not statistically significantly different from zero at the 10% significance level. These results suggest that the differences between treatment and control groups in the periods before Medicaid expansion are not statistically significantly different from zero, conditional on hospital and year fixed effects, which further supports our critical assumption that the treatment and control groups have parallel trends before the treatment. The coefficients of *Medicaid Expansion* are similar to those from our main analysis and statistically significantly different from zero, suggesting a sharp and significant effect of Medicaid expansion.

5.3.2. Posttreatment Effect. To analyze how the effect of Medicaid expansion changes over time, we first modify the difference-in-differences model by including a vector of posttreatment dummies. More specifically, we denote by $Medicaid_{it}^{k}$, where $k = \{0, 1, 2\}$, a vector of binary variables, each of which equals one if the state of hospital i expanded Medicaid k years before year t and zero otherwise. Comparing coefficients of $Medicaid_{it}^{0}$, $Medicaid_{it}^{1}$, and $Medicaid_{it}^{2}$ allows us to understand whether the effect of Medicaid expansion fades out, stays constant, or increases over time.

We then modify the difference-in-differences model by including a vector of both pretreatment and posttreatment dummies, as described earlier (i.e., $Medicaid_{it}^{l}$, where $l = \{-1, 0, 1, 2\}$). This comprehensive model allows us not only to estimate the pretreatment and posttreatment effects simultaneously, but also to understand whether the insignificance of pretreatment dummies in the first robustness check is due to a lack of statistical power.

The results from these two models are summarized in Tables E2 and E3 in the online appendix. We see that the coefficients of $Medicaid_{it}^{1}$ are larger than those of $Medicaid_{it}^{0}$ and similar to those of $Medicaid_{it}^{2}$, which suggests that the effect of Medicaid expansion increases in the first year after the expansion and stabilizes afterward. From Table E3 in the online appendix, we see that the coefficients of posttreatment dummies are statistically significantly different from zero, but those of the pretreatment dummy are not. These results suggest that the insignificance of the pretreatment effect is not due to a lack of statistical power.

5.3.3. Propensity Score Matching. To address potential concerns that the treatment and control groups have different characteristics, we use propensity score matching to select a sample of hospitals with similar pre-expansion characteristics and outcomes. This approach leads to 540 hospitals in both the treatment and control groups.[9] Table E4 in the online appendix summarizes the characteristics of these two groups one year before Medicaid expansion. We use t-tests to check whether the matched samples are balanced. From the last column of Table E4 in the online appendix, we see that none of the differences between these two groups are statistically significant at the 10% significance level.

We then use the matched samples to estimate the effect of Medicaid expansion on ED wait time. From Table E5 in the online appendix, we see that the coefficients of *Medicaid Expansion* are positive and statistically significantly different from zero at the 10% significance level, suggesting that Medicaid expansion increases ED wait time. Interestingly, the coefficients estimated based on matched samples are similar to those estimated based on unmatched samples, suggesting that differences between the treatment and control groups in some of the pre-expansion characteristics are not a major concern in our study.

5.3.4. Potential Endogeneity Issues. Although propensity score matching addresses potential issues of selection based on observable characteristics, unobservable hospital or population characteristics may affect both ED wait time and a state's decision to adopt Medicaid expansion. Note that we are not concerned with *time-invariant* unobservable characteristics, because they are captured by hospital fixed effects in Equation (1). Our challenge is to address potential endogeneity issues caused by *time-variant* unobservable characteristics.

To overcome this challenge, we follow existing literature (see, e.g., Brot-Goldberg et al. 2017, Bapna et al. 2018, and Sun et al. 2020) to use early adopters as a control group and late adopters as a treatment group.[10] The main idea behind this approach is that because both early and late adopters eventually chose to expand Medicaid, using early adopters as a control group captures not only observable characteristics, but also some time-variant unobservable characteristics that affect a state's decision to expand Medicaid.

In our study, the control group includes the states that adopted Medicaid expansion in 2014, and the treatment group includes the states that adopted Medicaid expansion in 2015 or 2016. The results based on these newly defined treatment and control groups are summarized in Table E6 in the online appendix. We see that the coefficients of *Medicaid Expansion* are positive and statistically significantly different from zero, which further supports our conclusion that Medicaid expansion increases ED wait time. Note that the results from this robustness check are not statistically significantly different from those from the main analysis at the 10% significance level.

5.3.5. Alternative Models. In the main analysis, we use a fixed-effects model to analyze 2,236 hospitals across five years. In panels with few time periods and many subjects (commonly known as "small T and large N" panels), coefficients may be biased when subject fixed effects are included (Buddelmeyer et al. 2008, Hill et al. 2020). To address this concern, we use a random-effects model and the generalized method of moments (GMM) to re-estimate the effect of Medicaid expansion.

Table E7 in the online appendix summarizes the results from the random-effects model and the difference GMM (Arellano and Bond 1991). We see that the coefficients of *Medicaid Expansion* are positive and statistically significantly different from zero at the 1% significance level, suggesting that Medicaid expansion increases ED wait time. These coefficients are not statistically significantly different from each other or those in the main analysis at the 10% significance level, suggesting that our results in the main analysis are not driven by the fixed-effects model.

5.3.6. Alternative Sample Definitions. We use four alternative definitions of samples to check the robustness of our results. First, we use all states that expanded Medicaid between 2014 and 2016 as a treatment group. Second, we exclude the Delaware, District of Columbia, Massachusetts, New York, and Vermont because these states expanded Medicaid or similar coverage to childless adults with incomes up to 100% of the federal poverty line or greater between 2010 and 2013 (Miller and Wherry 2017). Third, we further exclude California, Connecticut, Hawaii, Minnesota, and Wisconsin because these states had substantial expansions of Medicaid between 2009 and 2010 (Black et al. 2019). Finally, we re-estimate the effect of Medicaid expansion by leaving one state out at a time.

The results from these four alternative sample definitions are summarized in Tables E8, E9, E10, and E11 in the online appendix. We see that all coefficients of *Medicaid Expansion* are positive and statistically significantly different from zero at the 10% significance level, suggesting that Medicaid expansion increases ED wait time. These coefficients are not statistically significantly different from each other or those in the main analysis at the 10% significance level, suggesting that our results in the main analysis are not driven by a particular state or a group of states. We note that the coefficients in the main analysis are more significantly different from zero than those in Table E10 are, possibly because the main analysis has a larger sample size than this robustness check (11,180 versus 8,500).

5.3.7. Impact of Health Exchanges. When the ACA was signed into law in 2010, it called for the creation of an exchange (also called "marketplace") in each state by October 1, 2013, the starting date of open enrollment in an exchange. The implementation of these exchanges will not affect our results from the difference-in-differences analysis if both the treatment and control groups experience a common "shock." However, states varied in their implementation of these exchanges. In particular, whereas 26 states defaulted to the federal exchange, 18 states created state-operated exchanges, and seven states established state-federal partnership exchanges (Kaiser Family Foundation 2014). These variations raise a concern that our estimated effects could be attributed to the differences in the exchanges implemented by different states.

We address this concern by analyzing the effect of different exchanges on the treatment and control groups.[11] More specifically, we first focus on the states that expanded Medicaid in 2014 and use those that defaulted to the federal exchanges as a control group and those that created state-operated exchanges as a treatment group to estimate the effect of state-operated exchanges. Similarly, we estimate the effect of state-federal partnership exchanges by using the states that defaulted to the federal exchanges as a control group and those that established state-federal partnership exchanges as a treatment group. We then perform similar analyses by focusing on the states that did not expand Medicaid between 2012 and 2016.[12]

The results from this robustness check are summarized in Tables E12, E13, and E14 in the online appendix. We see that the coefficients of *State-Operated Exchange* and *State-Federal Partnership* are not statistically significantly different from zero at the 10% significance level, which suggests that the effects of these two types of exchanges are not statistically significantly different from the effect of federal exchange. Therefore, we can treat these exchanges as a common shock experienced by all states and conclude that the results from our main analysis are not driven by the implementation of different exchanges.

5.3.8. States' Political Party Affiliation. In the main analysis, we do not include states' political party affiliation (i.e., Democrat versus Republican) as a hospital or population characteristic because a state's political party affiliation remains the same during our study period, and the systematic difference across hospitals can be captured by hospital fixed effects in the difference-in-differences model. However, if the political party affiliation affects both a state's decision to expand Medicaid and ED wait time, our difference-in-differences approach may lead to a biased estimate of the average treatment effect.

To address this concern, we first use descriptive statistics to compare the status of Medicaid expansion and ED wait time between Democrat- and Republican-controlled states, where a state's political party affiliation is determined based on the 2012 election results (The New York Times 2012). The proportion of hospitals with Medicaid expansion is 15.47% (standard deviation of 36.18%) in Republican-controlled states and 76.39% (standard deviation of 42.49%) in Democrat-controlled states, which suggests that Democrat-controlled states are more likely to expand Medicaid. The average pre-expansion (i.e., 2013) ED wait time is 31.45 minutes (standard deviation of 18.29) in

Republican-controlled states and 31.22 minutes (standard deviation of 18.53) in Democrat-controlled states, which suggests that states political party affiliation does not significantly affect ED wait time.

We then include states' political party affiliation as a characteristic for propensity score matching. This approach leads to 387 hospitals in both the treatment and control groups. Table E15 in the online appendix summarizes the characteristics of these two groups one year before Medicaid expansion. We see that none of the differences between these two groups are statistically significant at the 10% significance level. In particular, the proportion of hospitals in Democrat-controlled states is 57% (standard deviation of 50%) in the expansion group and 60% (standard deviation of 49%) in the nonexpansion group. Table E16 in the online appendix summarizes the results based on the matched samples. We see that the coefficients of *Medicaid Expansion* are similar to those estimated based on unmatched samples, which suggests that differences between the treatment and control groups in states' political party affiliation are not a major concern in our study.

Finally, we follow Keele et al. (2019) to perform sensitivity analyses by applying the bracketing bounds approach. Because this approach accounts for an unmeasured confounder that has a different effect in the posttreatment period, it addresses a potential concern that an event besides the treatment occurs at the same time and affects the treatment group in a differential fashion. To apply this approach, we first partition the control group into two subgroups: (1) Republican-controlled states and (2) Democrat-controlled states. We then use each subgroup as a new control group to estimate the effect of Medicaid expansion. From Table E17 in the online appendix, we see that both coefficients of *Medicaid Expansion* are positive and statistically significantly different from zero at the 5% significance level. These coefficients are not statistically significantly different from each other or those in the main analysis at the 10% significance level, which further suggests that our results are not driven by states' political party affiliation.

5.3.9. Alternative Outcome Measures. In this robustness check, we use the log-transformed boarding time and ED length of stay as alternative outcome measures and re-estimate the treatment effect. Because ED length of stay varies widely between patients who are admitted to an inpatient department and those who are discharged to go home we create two subcategories (designated as "inpatient ED length of stay" and "outpatient ED length of stay") and analyze them separately.

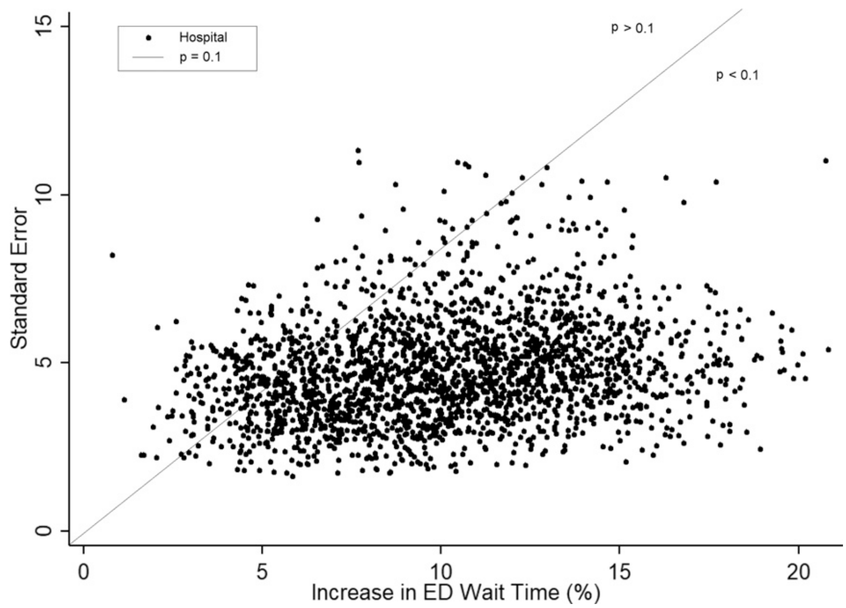
The results from this robustness check are summarized in Tables E18, E19, and E20 in the online appendix. We see that the coefficients across different models are positive and statistically significantly different from zero at the 10% significance level, suggesting that Medicaid expansion increases boarding time and ED length of stay. More specifically, the expansion increases boarding time by 3.4%, inpatient ED length of stay by 2.2%, and outpatient ED length of stay by 3.1%.

5.4. Heterogeneous Treatment Effects

We now use the FDCF approach described in Section 4.2 to estimate the heterogeneous effects of Medicaid expansion. To do so, we designate 2012 and 2013 as the pre-expansion period and 2015 and 2016 as the postexpansion period.[13] For a hospital (denoted by i), we first calculate its pre-expansion outcome using $Y_{i0} = [ln(EDWaitTime_{i,2012}) + ln(EDWaitTime_{i,2013})]/2$ and postexpansion outcome using $Y_{i1} = [ln(EDWaitTime_{i,2015}) + ln(EDWaitTime_{i,2016})]/2$. We then use the difference between pre-expansion and postexpansion outcomes (i.e., $Y_i = Y_{i1} - Y_{i0}$) as the new outcome variable. Recall that the treatment is Medicaid expansion (i.e., $W_i = Medicaid_i$), and features are pre-expansion hospital and population characteristics (i.e., $X_i = (Characteristics_{i,2012} + Characteristics_{i,2013})/2$). We use the local centering approach to regress out the effect of the characteristics on the outcome and treatment, so the forest focuses its splits on the features that affect the treatment effect instead of main outcome.

The results from heterogeneous treatment-effect analysis are summarized in Figure 2, where the horizontal axis indicates the estimated treatment effects across all hospitals and the vertical axis indicates the corresponding standard errors. From the spread of the scatterplot, we see a wide variation in the effect of Medicaid expansion with the smallest and largest values equal to 0.8% and 20.8%, respectively. The average effect across all hospitals is 10.4% (standard error of 3.8%), which is not statistically significantly different from the result from average effect analysis at the 10% significance level. Following the discussion in Abadie (2005), we note that the classical difference-in-differences estimate and the average treatment-effect estimate implied by the forest are two different estimators for the same parameter. They are justified by the same general identification strategy, but under different modeling assumptions. The lower-right side of the plot (i.e., the area below the oblique line) shows the effects that are statistically significantly different from zero at the 10% significance level. The smallest and largest values equal 3.1% and 20.8%, respectively. Comparing the upper-left and lower-right sides of the plot, we see that Medicaid expansion has a significant effect on a majority of hospitals (85%), but an insignificant effect on the other hospitals (15%).

Figure 2. Distribution of the Effects of Medicaid Expansion



Notes. The figure summarizes the results from heterogeneous treatment-effect analysis based on the 2012–2016 data from Hospital Compare, AHA, and ACS. The horizontal axis indicates the estimated treatment effects across different hospitals, and the vertical axis indicates the corresponding standard errors. The oblique line corresponds to $p = 0.1$.

We use a hypothesis test to analyze whether our proposed approach has succeeded in finding significant treatment-effect heterogeneity. The null hypothesis is that the effect of Medicaid expansion is homogeneous. We reject the null hypothesis if the effect is heterogeneous across two hospitals or two subgroups of hospitals. To test this hypothesis, we follow Athey and Wager (2019) to categorize hospitals into three subgroups (designated as "high," "medium," and "low") of equal size based on their estimated treatment effects. As noted in Athey and Wager (2019), this approach can provide insights about the strength of heterogeneity because the subgroups thus defined rely on using *out-of-bag* predictions and do not directly depend on the outcomes or treatments themselves. In our example, the average treatment effect of the low group is 6.58% (standard error of 4.40%), and that of the high group is 17.03% (standard error of 5.54%). The difference between these two groups is statistically significantly different from zero at the 10% significance level. Therefore, we reject the null hypothesis and conclude that our approach is able to find significant heterogeneity.

To understand what types of hospitals are more likely to see an increase in ED wait time after Medicaid expansion, we use the best linear projection approach (see, e.g., Athey and Wager 2019 and Semenova and Chernozhukov 2021) by projecting the treatment-effect function onto selected features. Note that this approach regresses doubly robust scores instead of estimated treatment effects on the features. Table 3 summarizes the results for the scenario in which the selected features are hospital characteristics. For ease of interpretation, we do not include interactions of different hospital characteristics. From the table, we see that the effect of Medicaid expansion is larger for teaching hospitals and hospitals with more beds and smaller for hospitals with more physicians.

Table 3. Best Linear Projection of the Conditional Average Treatment Effect

Hospital characteristics	Coefficient	Standard error
Teaching (Yes)	0.137*	0.077
Rural (Yes)	−0.019	0.061
Rural (Unknown)	−0.148	0.214
EHR Implementation (Partial)	0.073	0.070
EHR Implementation (Full)	0.030	0.063
EHR Implementation (Unknown)	0.124	0.130
Freestanding ED (Yes)	−0.013	0.113
Freestanding ED (Unknown)	−0.023	0.053
HMO (Yes)	−0.059	0.108
HMO (Unknown)	0.134	0.268
PPO (Yes)	0.115	0.114
ln(*Number of Beds*)	0.092***	0.030
ln(*Number of Physicians*)	−0.026*	0.014
Constant	−0.497**	0.195
Number of hospitals	2,236	

Notes. This table summarizes the results from the best linear project approach (see, e.g., Athey and Wager 2019 and Semenova and Chernozhukov 2021), where the dependent variable is the estimated treatment effects and the independent variable is hospital characteristics. The results are based on the 2012–2016 data from Hospital Compare, AHA, and ACS. Standard errors are clustered by state.

*$p \leq 0.1$; **$p \leq 0.05$; ***$p \leq 0.01$.

The effect of Medicaid expansion is larger for large teaching hospitals for three reasons. First, large teaching hospitals are more likely to be safety-net hospitals, which organize and deliver a significant level of healthcare and other health-related services to patients with no insurance or with Medicaid (Agency for Healthcare Research and Quality 2014). Second, large teaching hospitals have better quality of care (Ayanian and Weissman 2002, Dimick et al. 2004). Because patients consider care quality when choosing a healthcare provider (Smith et al. 2018), large teaching hospitals are more likely to see an increase in patient volume after Medicaid expansion. Finally, large teaching hospitals have higher utilization (Pines et al. 2012). Because the marginal effect of utilization on wait time increases as utilization increases (Green et al. 2006), large teaching hospitals are likely to see a larger increase in wait time for the same amount of increase in utilization.

The effect of Medicaid expansion is smaller for hospitals with more physicians for three reasons. First, the marginal effect of utilization on wait time decreases as the number of physicians increases (Green et al. 2006), so hospitals with more physicians are likely to see a smaller increase in wait time for the same amount of increase in utilization. Second, hospitals with more physicians have more flexibility in pooling patient queues to reduce wait time (Song et al. 2015) and matching physicians with patients to reduce treatment time (Wang and Pourghannad 2020). Third, because physicians may overcome high utilization by speeding up (KC 2013, Berry Jaeker and Tucker 2016), hospitals with more physicians are more flexible in speeding up the hospital admission and discharge process in the face of increased demand.

Interestingly, we do not find the effect of Medicaid expansion on wait time to be different between rural and urban hospitals. Because rural areas have a larger number of uninsured patients than urban areas (Kaiser Family Foundation 2013b), we can reasonably expect rural hospitals to have a larger increase in utilization. However, as discussed earlier, the marginal effect of utilization on wait time increases as utilization increases. Because rural hospitals have lower utilization than urban hospitals before Medicaid expansion (Greenwood-Ericksen and Kocher 2019), a larger increase in utilization at rural hospitals after Medicaid expansion may lead to a similar increase in wait time at urban hospitals. Also, existing studies (see, e.g., Lindrooth et al. 2018) find that Medicaid expansion reduces rural hospital closures. The reduction in hospital closure helps divert newly enrolled Medicaid patients to more hospitals and therefore mitigates the effect of increased demand.

The results from average effect analysis are useful to policymakers, especially those in nonexpansion states, in helping them decide whether to expand Medicaid. However, they are not so useful to individual hospital managers, because an average effect of 10.4% does not mean that all hospitals will see a 10.4% increase in wait time after Medicaid expansion. Our heterogeneous treatment-effect analysis shows that the effect varies widely across hospitals, with the smallest and largest values equal to 3.1% and 20.8%, respectively. More specifically, we find the effect to be larger for large teaching hospitals and smaller for hospitals with more physicians. These results can help hospital managers understand the effect of Medicaid expansion on their own hospitals, instead of all hospitals as a group. They also allow hospital managers to more accurately estimate the amount of additional resources (e.g., number of physicians and hospital beds) needed to mitigate the potential negative effect of Medicaid expansion on wait time.

6. Conclusion

Medicaid expansion and ED crowding are two of the most commonly discussed topics in American society. Since its implementation in 2014, Medicaid expansion has helped millions of uninsured patients obtain insurance coverage. Intuitively, an increase in the number of insured patients increases the number of ED visits and, therefore, crowding. However, the reverse may be true if Medicaid expansion shifts some of the ED volume to other departments, such as inpatient care. The complex relationship between Medicaid expansion and ED crowding has drawn considerable attention but, unfortunately, has not been well studied in the existing literature, possibly due to a lack of data or suitable methodology.

We use the recently released Hospital Compare data to study the effect of Medicaid expansion on ED wait time using the difference-in-differences approach, where the treatment group includes the states that expanded Medicaid at the beginning of 2014 and the control group includes the states that did not expand Medicaid during our study period. By comparing the treatment and control groups before and after Medicaid expansion, we find that Medicaid expansion increases ED wait time by 10.4%. Our results are robust to alternative sample definitions, potential endogeneity issues, and the impact of health changes. To understand the mechanisms behind our results, we compare the total number of ED visits, hospital beds, physicians, and nurses before and after Medicaid expansion (see Online Appendix H). We find that an increase in ED wait time is due not to a relative decrease in the number of hospitals beds, physicians, or nurses, but, rather, to a relative increase in the total number of ED visits spurred by the change in insurance coverage.

In addition to estimating the average effect of Medicaid expansion, we develop an FDCF approach by incorporating the first-difference approach into a causal forest approach. We conduct comprehensive simulation studies to assess the performance of the FDCF approach and find that it has smaller MSEs than direct applications of the causal forest approach. By applying this approach to our empirical example, we find that the effect of Medicaid expansion varies widely across different hospitals. More specifically, Medicaid expansion has a significant effect on 85% of hospitals, but an insignificant effect on the other 15% of hospitals. The effect is larger for teaching hospitals and hospitals with more beds, but smaller for hospitals with more physicians. The results of this study are useful to policymakers and hospital managers understanding the average and heterogeneous effects of Medicaid expansion.

Endnotes

[1] The data set description is available at https://data.cms.gov/provider-data/dataset/yv7e-xc69.

[2] The quarterly updates are based on the data submitted by hospitals in the past 12 months. To avoid potential issues with overlapping data, we use yearly instead of quarterly data to analyze the effect of Medicaid expansion.

[3] The data set description is available at https://www.census.gov/programs-surveys/acs/about.html.

[4] The data set description is available at https://www.aha.org/other-resources/2018-01-08-aha-data-and-directories.

[5] Out of these 191 hospitals, 168 hospitals are from the states that either expanded Medicaid at the beginning of 2014 or did not expand Medicaid before 2016. In Online Appendix A, Table A1 compares the hospitals that are included and excluded from this study. Table A2 in the online appendix compares the excluded hospitals in the treatment and control groups one year before the expansion.

[6] The 24 states in the treatment group are Arizona, Arkansas, California, Colorado, Connecticut, Delaware, Hawaii, Illinois, Iowa, Kentucky, Maryland, Massachusetts, Minnesota, Nevada, New Jersey, New Mexico, New York, North Dakota, Ohio, Oregon, Rhode Island, Vermont, Washington, and West Virginia, plus the District of Columbia. The 20 states in the control group are Alabama, Florida, Georgia, Idaho, Kansas, Louisiana, Maine, Mississippi, Missouri, Nebraska, North Carolina, Oklahoma, South Carolina, South Dakota, Tennessee, Texas, Utah, Virginia, Wisconsin, and Wyoming.

[7] Table D1 in Online Appendix D compares the hospital and population characteristics of the two groups.

[8] The parallel-trends assumption may be violated due to the "woodwork effect" if the expansion states have more individuals coming out of the woodwork than the nonexpansion states. We address this concern by first estimating the number of previously eligible individuals and then including additional independent variables to control for the woodwork effect (see Online Appendix F for more details).

[9] We use a caliper of 0.1 standard deviations of the propensity score. Using alternative caliper values (e.g., 0.05 or 0.2) does not change the main conclusion of this robustness check.

[10] As an additional analysis, we follow Rosenbaum and Silber (2009) by performing sensitivity analyses to determine how strong the effects of the unobservable features would have to be to change the substantive conclusion from our study (see Online Appendix G for more details).

[11] The main idea behind this approach is that if different exchanges have the same effect on ED wait time, we can treat them as a common shock to both the treatment and control groups.

[12] None of these states created state-operated exchanges, so we estimate the effect of state-federal partnership exchanges only.

[13] Using alternative pre-expansion and postexpansion periods (e.g., 2013 versus 2015 or 2012 versus 2016) does not change the main conclusion of this analysis.

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